STRING HANDLING

String:

A sequence of characters is known as string.

Character can be a letter, digit, whitespace or any other symbol.

There are two types of string literals:

1) Single line string literals

2) Multi line string literals

1) Single line string literals:

Single line string literal can be in single quotes, double quotes, triple single quotes & triple double quotes.

Examples:

1) 'welcome'

2) "weclome"

3) '''welcome'''

4) """welcome"""

2) Multi line string literals:

Multi line string literal can be in triple single quotes & triple double quotes.

Examples:

1) '''welcome

    to

    python'''

2) """welcome

    to

python"""

String supports positive indexing & negative indexing.

Positive index starts with 0 from left to right and negative index starts with -1 from right to left.

Example:

```
s="welcome"
print(s[3])
print(s[-2])
```

Traversing a string by using while or for loop.

Examples:

1)

```
s="welcome"
i=1
while i<7:
   print(s[i])
   i+=1
```

2)

```
s="welcome"
for i in s:
   print(i)
```

String supports slicing operation also.

A sub string of a stirng is called a slice.

Example:

```
s="welcome"
```

```
print(s[3:5])
print(s[-4:-1])
print(s[:4])
print(s[2:])
print(s[::3])
print(s[::-1])
```

Concatenating two string by using + operator.

Example:

```
s1="Taj"
s2="Mahal"
s3=s1+s2
print(s3)
```

Multiplying a string with numbers will print the string in specified number of times.

Example:

```
s="hello"
print(s*100)
```

Strings are immutable objects.

Immutable object means the value of an object cannot be changed.

If you try to change existed value then new object is created.

Memory space can be saved by using immutable objects.

By using del keyword we can delete object.

String handling functions & methods:

1) isalnum() => It returns true if string has at least 1 character and every character is either a number or an alphabet, otherwise it returns false

2) isalpha() => It returns true if string has at least 1 character and every character is an alphabet, otherwise returns false.

3) isdigit() => It returns true if string contains only digits, otherwise returns false.

4) isspace() => It returns true if string contains only whitespace characters, otherwise returns false.

5) islower() => It returns true if string contains atleast 1 character and every character is a lowercase alphabet, otherwise returns false.

6) isupper() => It returns true if string contains atleast 1 character and every character is a uppercase alphabet, otherwise returns false.

7) capitalize() => It is used to capitalize first letter of the string.

8) lower() => It converts all characters in the string into lowercase.

9) upper() => It converts all characters in the string into uppercase.

10) swapcase() => Toggles the case of every character(Uppercase character becomes lowercase and lowercase character becomes uppercase)

11) title() => It is used to capitalize each word first letter in a string.

12) count(str) => It counts the number of times str occurs in a given string

13) startswith(prefix) => It returns true if the given string starts with the specified prefix, otherwise returns false.

14) endswith(suffix) => It returns true if the given string ends with the specified suffix, otherwise returns false.

15) find(str) => It returns the index of str in a given string if the str present, otherwise it returns -1

16) index(str) => It returns the index of str in a given string if the str present, otherwise it raises an exception.

17) len(string) => It returns the length of the specified string.

18) max(string) => It returns the highest ascii value alphabet.

19) min(string) => It returns the lowest ascii value alphabet.

20) strip() => It removes all leading and trailing whitespaces in a given string.

By

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